

Lab (CL-2005)**Date:** Tuesday, March 31st, 2026**Time:** 1:30 PM to 4:00 PM**Lab Instructor:** Miss Saira Ayub**Total Time (mins):** 120**Total Weightage:** 25**Total Marks:** 50**Total Questions:** 03**Paper Type:** A

Section

Signature

Do not write below this line

Attempt all the questions!

Important Instructions:

- Return the question paper mention your student ID on it.
- Read each question completely before answering it. There are 3 questions on 3 pages.
- Attempt all the given questions. All questions are carrying different points.
- Cheating in any case will lead to F-GRADE as per university rule.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.

Question 1: LLO-04 Database Design & Manipulation (Work-Sphere Management) [Marks: 14]**Time: 40 Minutes]**

"WorkSphere" is a co-working platform that offers multiple types of workspaces, each defined by its category, capacity, and hourly rate. Users register as members by providing their personal and professional details, after which they can reserve workspaces for specific dates and time slots. The system ensures that a workspace cannot be double-booked during overlapping timings.

Members can optionally add extra services such as internet, refreshments, or printing to their reservations, which increases the total cost. For every confirmed reservation, a financial record is generated to track the total amount, payment status, and transaction date. The system also maintains data integrity by handling reservations linked to unavailable workspaces and preserving relationships between all records.

1. Identify entities and their attributes from the above scenario. Apply appropriate and necessary constraints.
2. Add at least three (03) records in each table to make the database operational.
3. Retrieve the names of members who registered before January 2024 along with the total number of reservations they have made, including those with zero reservations.
4. Increase the hourly rate of all "Meeting Room" workspaces by 15%, only if their current rate is below the overall average workspace rate.
5. Delete all reservations that reference workspaces which no longer exist in the system.
6. Display members who have used more than one distinct service across their reservations, along with the count of those services.
7. Find the workspace that has generated the highest total revenue based on all associated financial records.

National University of Computer and Emerging Sciences
Karachi Campus

Question 2: LLO-03 Intermediate Queries on the "Chinook" Database [Marks: 10 | Time: 20 Minutes]

Import the Chinook schema. Execute the following queries. Each query contains [2 marks]

1. Calculate the average total amount of all invoices, rounded to two decimal places, but only include those invoices whose total is greater than the minimum invoice total in the database.
2. Display the full name of each customer by combining their first and last names, along with their city and country merged into a single column labeled "Location". Sort the result based on country.
3. Display the name and unit price of all tracks whose price falls between 0.99 and 1.99, and also calculate the total cost if 10 units of each track were purchased. Sort the result by unit price in descending order.
4. Retrieve the full names of employees along with their calculated age, and include the total number of customers assigned to each employee. Ensure that all employees are displayed, even if they are not assigned to any customers, by properly using joins.
5. Generate a detailed report showing each track name, its corresponding album title, and the artist name. Additionally, include only those tracks whose names exceed 20 characters in length, and present the track names in uppercase format. The result should demonstrate proper use of joins across multiple tables.

Question 3: LLO-03 Execute the following high-complexity tasks on the Chinook database. [Marks: 26 | Time: 55 Minutes]

1. List all tracks showing their names and duration in milliseconds, but only include those tracks whose duration is longer than the average duration of tracks within the same media type category, so that comparison is done relative to their specific media type.
2. Provide a list of all employees showing each employee's last name along with the last name of the employee who acts as their immediate supervisor or manager, capturing the hierarchical reporting structure within the organization.
3. Determine which tracks exist in the system but have never been sold, meaning tracks that are recorded in the track collection but do not appear in any sales records, and present their unique identifiers.
4. Generate a list that includes the names of artists, the titles of albums, and the names of tracks, but only for tracks that belong to the genre "Jazz," ensuring that each track is properly associated with its corresponding album and artist.
5. Create a comprehensive collection of information that combines track names, album titles, genre names, and media formats, resulting in a detailed view of all tracks in the system with their associated album, genre, and media type.
6. From the comprehensive track information, calculate the total number of tracks that are in the media format "MPEG audio file" and are performed by the artist named "Queen," counting only those tracks that satisfy both conditions.
7. Identify all invoices where the total amount is greater than the average total of all invoices that have been processed by the employee named "Jane Peacock," ensuring that only invoices exceeding this employee-specific average are included.
8. Using the collection of high-value invoices, compute for each city the sum of all invoice totals and display the name of each billing city along with the total sales amount associated with that city, providing insight into city-level revenue.